Math Inquirers

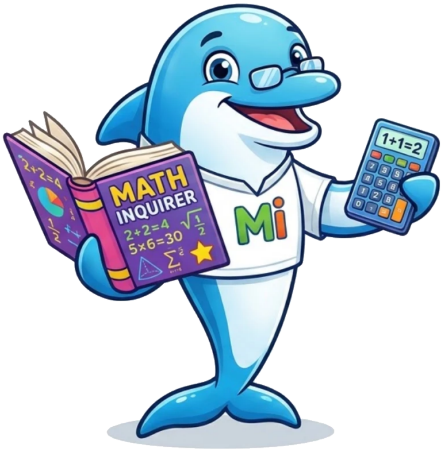
Think. Question. Prove.

Level 1 (ages 6-8) - Term 2

Tutor Answer Key

Book 3

Lessons 11 to 15



FOR THE TUTOR AND GROWN-UPS ONLY - keep away from the child

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Math Inquirers Level 1, Term 2 - Tutor Answer Key, Book 3

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Answer key: Lesson 1, Who Is Where?

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Study the picture. The door is on the left.

Answer: True, False, True

Ali is 1st, Kumar is 2nd, Wei is 3rd, Jenny is 4th and Lily is 5th. So 'Wei is 3rd' is true. 'Jenny is 5th' is false, because Jenny is 4th. Lily is the last child, at the back of the line: true.

Watch for: Counting from the wrong end.; Saying Jenny is 5th because Lily was missed.

S2. Fill in the places of the children. The door is on the left.

Answer: 1st, 2nd, 3rd, 4th, 5th, 6th, 7th

The child next to the door is 1st. Then 2nd, 3rd, 4th, 5th, 6th and 7th for the last child.

Watch for: Writing 2th or 3th.; Counting from the wrong end.; Writing 1 instead of 1st.

S3. Count the children. The door is on the left.

Answer: 10 children; the last child is 10th

Count 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. There are 10 children. The last child is 10th.

Watch for: Writing 10 for both blanks.; Miscounting a child twice.

S4. Aisha is 2nd in the line. Kumar stands right behind Aisha.

Answer: Kumar is 3rd

Aisha is 2nd. The very next place is 3rd, so Kumar is 3rd.

Watch for: Writing 1st because Kumar is behind.; Writing 2nd, Aisha's own place.

S5. Jenny is 2nd from the front in a line of 6 children.

Answer: 5th from the back

Count from the back: the last child is 1st, then 2nd, 3rd, 4th, and Jenny is 5th from the back. Check: the places from the front and the back are 2 and 5 and they make 7, one more than 6.

Watch for: Writing 2nd again.; Writing 4th by counting only the children behind Jenny.

S6. Wei is 1st. Lily is 3rd. Kumar stands between Wei and Lily.

Answer: Kumar is 2nd

Wei is 1st and Lily is 3rd. The place between them is 2nd, so Kumar is 2nd.

Watch for: Writing 4th.; Writing 1st or 3rd, the place of one of the clues.

S7. Ben is 5th in the line.

Answer: 4 children

The children in front of Ben are 1st, 2nd, 3rd and 4th. That is 4 children. Ben himself is 5th, and he is not in front of himself.

Watch for: Writing 5, Ben's own place.; Writing 6 by counting Ben and the child behind.

Think harder (page 3)

T1. Mei is 5th from the front and 5th from the back.

Answer: No; 9 children

In front of Mei are 4 children. Behind Mei are 4 children. Together with Mei that is 4 + 1 + 4 = 9 children. Kumar's 10 counted Mei twice (5 + 5).

Watch for: Adding 5 + 5 = 10 and counting Mei twice.; Forgetting Mei and writing 8.

T2. Ali is 3rd and Mei is 9th in a line of 10 children.

Answer: 5 children

The children between Ali and Mei are 4th, 5th, 6th, 7th and 8th. That is 5 children. Ali and Mei are not between themselves.

Watch for: Writing 6 by doing 9 - 3.; Counting Ali and Mei as well and writing 7.

T3. The door is on the right. Count the places from the door.

Answer: 6th

The door is on the right, so the child next to the door is 1st. Counting along to the left, Jason is the 6th child. Check: he is 2nd from the left end, and 7 - 2 + 1 = 6.

Watch for: Counting from the left end and writing 2nd.; Writing 5th by missing one child.

T4. There are 8 children in a line. Wei is 2nd from the front. Lily is 3rd from the back.

Answer: 3 children

Lily is 3rd from the back, so she is 6th from the front (8 children: 8 - 3 + 1 = 6). Wei is 2nd. Between them are the 3rd, 4th and 5th: 3 children.

Watch for: Writing 4 by doing 6 - 2.; Forgetting to turn Lily's place around.

Go further (page 4)

G1. Six children sit round a table. The seats are numbered 1 to 6. Ali is in seat 1, Mei is in seat 3 and Ben is in seat 5.

Answer: Seat 2; seat 4; seat 6

(a) Seat 2 is next to seat 1 and seat 3, so Siti sits in seat 2. (b) Opposite seat 1 is seat 4, three seats round. (c) Seat 6 is next to seat 5 and seat 1, so Kumar sits in seat 6.

Watch for: Putting Siti in seat 4 or seat 6, which is next to only one of them.; Choosing seat 3 as opposite seat 1.

G2. A line has 13 children. Mei is 4th from the front. Ben is 3rd from the back.

Answer: 6 children; 7th

(a) Ben is 11th from the front (13 - 3 + 1 = 11). Between the 4th and the 11th are the 5th to the 10th: 6 children. (b) The middle child has 6 children in front and 6 behind, so is 7th.

Watch for: Writing 7 for (a) by doing 11 - 4.; Writing 6th or 6 for the middle child.

G3. Four children ran a race. Siti was 1st. Ben beat Ali. Ali beat Mei.

Answer: Ben 2nd; Mei 4th

Siti is 1st. Ben beat Ali, and Ali beat Mei, so the order is Siti, Ben, Ali, Mei. Ben was 2nd and Mei was 4th.

Watch for: Putting Ali 2nd.; Writing 3rd for Mei.

Investigation (page 5): Two places for one child

For a line of 5, every child's two places add to 6: 1 and 5, 2 and 4, 3 and 3, 4 and 2, 5 and 1. For a line of 8 they add to 9. The two places always add to one more than the number of children, because the child is counted once from each end.

Look for: A child who writes both places for several children, adds them, and notices the total is the same for every child in one line, and one more than the number of children. Praise the notice, not the speed.

Exit ticket (page 5)

E1. Ben is 4th in the line.

Answer: 3 children

The children in front of Ben are the 1st, 2nd and 3rd. That is 3 children, because Ben himself is 4th.

Watch for: Writing 4, Ben's own place.

E2. Mei is 5th from the front in a line of 6 children.

Answer: 2nd from the back

Mei is 5th from the front, so only 1 child is behind her. Counting from the back: that child is 1st and Mei is 2nd.

Watch for: Writing 1st or 5th.

E3. Aisha is 3rd in the line. Kumar is right behind Aisha. Wei is right behind Kumar.

Answer: Wei is 5th

Aisha is 3rd. Kumar is right behind, so Kumar is 4th. Wei is right behind Kumar, so Wei is 5th.

Watch for: Writing 4th.

Sneak-peek puzzles (not in the workbook)

1. The dot next to the door is 1st. The 2nd dot is the one after it.
2. 4. Ben is 5th. In front of Ben are the 1st, the 2nd, the 3rd and Mei, who is 4th.

Answer key: Lesson 2, Bar Strips: More and Fewer

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Ali has 12 stickers. Mei has 8 stickers.

Answer: 4 more stickers

Ali's strip is longer. The extra part is the gap: 12 take away 8 is 4. Ali has 4 more stickers than Mei.

Watch for: Adding 12 + 8 = 20.; Writing 12, the whole strip.

S2. Ben has 15 marbles. Kumar has 7 more marbles than Ben.

Answer: 22 marbles

Kumar has Ben's 15 marbles and 7 more. 15 + 7 = 22. Kumar has 22 marbles.

Watch for: Taking away: 15 - 7 = 8.; Writing 7, only the extra part.

S3. Siti has 34 beads. Dev has 17 more beads than Siti.

Answer: Dev has 51 beads; 85 altogether

(a) Dev has Siti's 34 and 17 more: 34 + 17 = 51. (b) Altogether is both strips end to end: 34 + 51 = 85.

Watch for: Writing 17 for Dev.; Writing 51 for the total.

S4. Mei has 9 cards. Ravi has 5 cards. Draw a strip for Mei and a strip for Ravi.

Answer: 4 more cards (Mei's strip is longer by 4 cells)

Mei's strip has 9 boxes and Ravi's strip has 5 boxes. The extra 4 boxes on Mei's strip are the gap: 9 take away 5 is 4.

Watch for: Drawing both strips the same length.; Adding 9 + 5 = 14.

S5. Mei has 28 stickers. Ali has 6 more stickers than Mei.

Answer: Ali has 34 stickers; 62 altogether

(a) Ali has 28 and 6 more: 28 + 6 = 34. (b) Altogether: 28 + 34 = 62.

Watch for: Writing 22 by taking away.; Writing 34 for the total.

S6. Wei has 45 stamps. Jenny has 30 stamps.

Answer: 15 more stamps

Wei's strip is longer. The gap is 45 take away 30, which is 15. Wei has 15 more stamps than Jenny.

Watch for: Adding 45 + 30 = 75.; Writing 30.

S7. Aisha has 40 beads. Lily has 24 beads. Which number sentence finds how many more beads Aisha has than Lily? Write its number in the brackets.

Answer: (2) 40 - 24

Aisha's strip is longer, and we want the gap. The gap is the long strip take away the short strip: 40 - 24. That is sentence (2). Adding would join the strips.

Watch for: Choosing (1) because it uses both numbers.; Choosing (3) 40 + 40.

Think harder (page 3)

T1. Ali and Mei have 40 stickers altogether. Ali has 8 more stickers than Mei.

Answer: Mei 16; Ali 24

Take away Ali's extra 8: 40 - 8 = 32. Two equal strips share 32, so each is 16. Mei has 16. Ali has 16 and 8 more, which is 24. Check: 16 + 24 = 40.

Watch for: Halving 40 and writing 20 and 20.; Writing 32 and 8.

T2. Ali has 30 stickers. Mei has 18 stickers. Ali gives some stickers to Mei. Then they have the same number.

Answer: 6 stickers

The gap is 30 - 18 = 12. Ali gives half of the gap: 6. Then Ali has 24 and Mei has 24.

Watch for: Writing 12, the whole gap.; Writing 24, the number each ends with.

T3. Wei has 18 fewer marbles than Ravi. Ravi has 52 marbles.

Answer: Wei 34; 86 altogether

Wei has 18 fewer, so Wei's strip is Ravi's strip with 18 cells taken off: 52 - 18 = 34. Altogether: 52 + 34 = 86.

Watch for: Adding 52 + 18 = 70, because the words say 18 and 52.; Writing 70 for the total.

Go further (page 4)

G1. 52 books are on two shelves. 7 books move from the top shelf to the bottom shelf. Then both shelves have the same number of books.

Answer: Top 33; bottom 19

52 shared into two equal strips is 26 each. The top shelf lost 7 to get to 26, so it had 26 + 7 = 33. The bottom shelf gained 7, so it had 26 - 7 = 19. Check: 33 + 19 = 52.

Watch for: Writing 26 for the top shelf.; Writing 26 - 7 = 19 for the top shelf.

G2. Ali, Ben and Mei have 90 cards altogether. Ben has 10 more cards than Ali. Mei has 10 more cards than Ben.

Answer: Ali 20; Mei 40

Ben has 10 more than Ali, and Mei has 20 more than Ali. Take away the extra 10 and 20: 90 - 30 = 60. Three equal strips share 60: 20 each. Ali has 20, Ben has 30 and Mei has 40. Check: 20 + 30 + 40 = 90.

Watch for: Writing 30 for Ali (90 shared into three).; Forgetting Mei's extra 20.

G3. Kumar has 12 more marbles than Siti. Siti gives 4 marbles to Kumar.

Answer: 20 more marbles

At first the gap is 12. Kumar gains 4 and Siti loses 4, so the gap grows by 4 + 4 = 8. Now Kumar has 12 + 8 = 20 more marbles than Siti.

Watch for: Writing 16 (12 + 4).; Writing 8, the change in the gap only.

Investigation (page 5): Same total, different gaps

With 2 more: Ali 6 and Mei 4. With 4 more: Ali 7 and Mei 3. With 6 more: Ali 8 and Mei 2. Each time the bigger number goes up by 1 and the smaller goes down by 1, as the gap goes up by 2. With a total of 10 the gap is always an even number, and the bigger number is half of the total and half of the gap added together.

Look for: A child who draws the strips, finds each pair, checks that the pair adds to 10, and notices a pattern (bigger goes up by 1, smaller goes down by 1; the gap goes up by 2). Praise the notice, not the speed.

Exit ticket (page 5)

E1. Ali has 13 stickers. Mei has 9 stickers.

Answer: 4 more stickers

Ali's strip is longer. The gap is 13 take away 9, which is 4.

Watch for: Writing 22 by adding.

E2. Ben has 27 marbles. Siti has 15 more marbles than Ben.

Answer: 42 marbles

Siti has Ben's 27 and 15 more: 27 + 15 = 42.

Watch for: Writing 12 by taking away.

E3. Wei has 25 cards. Jenny has 8 more cards than Wei.

Answer: 58 cards

Jenny has 25 + 8 = 33 cards. Altogether: 25 + 33 = 58.

Watch for: Writing 33, only Jenny's cards.

Sneak-peek puzzles (not in the workbook)

1. 8. Ali has Mei's 5 and 3 more: 5 + 3 = 8.
2. 2. Together they have 24, so each has 12 at the end. Ali goes from 14 to 12, so he gives 2. (The gap of 4 is shared: half of 4 is 2.)

Answer key: Lesson 3, Doubles and Near Doubles

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the dominoes. Fill in the missing numbers.

Answer: 6, 8, 10

Each domino has two halves with the same number of dots. 3 + 3 = 6. 4 + 4 = 8. 5 + 5 = 10.

Watch for: Counting only one half of the domino.; Writing 5 + 5 = 55 (joining the digits).

S2. Mei has 6 red counters and 6 yellow counters. Fill in the missing numbers.

Answer: 6 + 6 = 12; double 6 is 12

The first frame has 6 red counters and the second has 6 yellow counters. 6 + 6 = 12, so double 6 is 12.

Watch for: Counting only one frame (6).; Writing 66.

S3. Fill in the missing numbers.

Answer: double 2 is 4; double 5 is 10; double 7 is 14; double 9 is 18

Double 2 is 2 + 2 = 4. Double 5 is 5 + 5 = 10. Double 7 is 7 + 7 = 14. Double 9 is 9 + 9 = 18.

Watch for: Writing the number again (double 7 is 7).; Adding 2 to the number (double 7 is 9).

S4. Look at the bars. Fill in the missing numbers.

Answer: 3 tens + 3 tens = 6 tens; 30 + 30 = 60; double 40 is 80

3 tens and 3 tens make 6 tens. 6 tens is 60. Double 4 tens is 8 tens, which is 80.

Watch for: Writing 6 for 30 + 30.; Writing 33 or 303.

S5. Ali has 8 stickers. Ben has 8 stickers. How many stickers do they have altogether?

Answer: 8 + 8 = 16; they have 16 stickers

Ali has 8 and Ben has 8. 8 + 8 = 16. They have 16 stickers altogether.

Watch for: Writing 8 (only one child).; Writing 88.

S6. Read each statement. Decide if it is True or False.

Answer: True, False, True, True

Double 4 is 8: True. Double 6 is 12, not 14: False. 9 + 9 is two of the same number, so it is a double: True. Double 30 is 60: True.

Watch for: Marking Double 6 is 14 as true because 6 + 6 is close to 14.; Saying 9 + 9 is not a double because 9 is a big number.

Think harder (page 3)

T1. Look at the counters. The ringed counter is 1 more. Fill in the missing numbers.

Answer: double 5 is 10; 5 + 6 = 11

5 red counters and 5 yellow counters make double 5 = 10. The ringed yellow counter is 1 more, so 5 + 6 = 10 + 1 = 11.

Watch for: Writing 10 for both parts.; Adding 2 instead of 1.

T2. Add.

Answer: 7, 15, 21, 5

3 + 4: double 3 is 6, and 1 more is 7. 8 + 7: double 7 is 14, and 1 more is 15. 10 + 11: double 10 is 20, and 1 more is 21. 2 + 3: double 2 is 4, and 1 more is 5.

Watch for: Taking 1 away instead of adding 1 after the smaller double.; Writing the double itself (6 for 3 + 4).

T3. Which sum is not a near double? Write the number in the bracket.

Answer: (3) 5 + 8

6 and 7 are next to each other. 8 and 9 are next to each other. 9 and 10 are next to each other. 5 and 8 are not next to each other (they are 3 apart), so 5 + 8 is not a near double.

Watch for: Choosing 9 + 10 because 10 looks different.; Looking at the total instead of the two numbers.

T4. Kumar adds 8 + 9. Step 1: 8 + 8 = 16. Step 2: 16 - 1 = 15. Answer (a) and (b).

Answer: No; 8 + 9 = 17

9 is 1 more than 8, so the total is 1 more than 16: 16 + 1 = 17. Kumar took 1 away instead of adding 1.

Watch for: Saying Yes because 16 - 1 looks like a good step.; Writing 15 for the right answer.

Go further (page 4)

G1. Add 8 + 9 + 8 + 7 + 8. Show your working on the lines.

Answer: 40

The 9 is 1 more than 8 and the 7 is 1 less than 8, so they make up for each other. The five numbers are the same as five 8s: 8 + 8 + 8 + 8 + 8 = 40. Another way: 8 + 8 = 16, 9 + 7 = 16, and 16 + 16 + 8 = 40.

Watch for: Adding from left to right and slipping.; Leaving one number out.

G2. Ben says 9 + 10 + 11 is the same as 10 + 10 + 10. Answer (a) and (b). Show your working on the lines.

Answer: Yes; 19 + 20 + 21 = 60

9 is 1 less than 10 and 11 is 1 more than 10, so they make up for each other: 9 + 10 + 11 = 30 = 10 + 10 + 10. In the same way 19 + 20 + 21 is three 20s, which is 60.

Watch for: Saying No because the numbers look different.; Writing 50 for the three numbers.

G3. Find double 35. Fill in the missing numbers. Show your working on the lines.

Answer: 60, 10, 70

35 is 30 and 5. Double 30 is 60. Double 5 is 10. 60 + 10 = 70. So double 35 is 70.

Watch for: Writing 610 (joining 60 and 10).; Forgetting to double the 5.

Investigation (page 5): Make two groups the same

It works for 6 and 8 (both 7), and for 5 and 9 (both 7): the two numbers are 2 apart or 4 apart and the total is a double (14). It does not work for 7 and 8: the total is 15 and one dot is always left over.

Look for: A child who tries several pairs, moves dots, and notices that when the total is a double the two groups can be made the same, and that 7 and 8 always leave one dot over. Credit any clear reason. Do not use the words odd or even yet: that is the next lesson.

Exit ticket (page 5)

E1. Fill in the missing number.

Answer: 18

9 + 9 = 18. Move 1 from one 9 to the other: 10 and 8.

Watch for: Writing 9 or 11.; Writing 99.

E2. Fill in the missing number.

Answer: 60

Double 3 is 6, so double 3 tens is 6 tens: 60.

Watch for: Writing 6.; Writing 33 or 303.

E3. Add.

Answer: 17

Double 8 is 16 and 1 more is 17. Or double 9 is 18 and 1 less is 17.

Watch for: Writing 18 (the double of 9).; Writing 16.

Sneak-peek puzzles (not in the workbook)

1. 8
2. Yes. 6 + 7 is 6 + 6 with one more counter, so 12 + 1 = 13.

Answer key: Lesson 4, Odd + Odd: Always, Sometimes, Never

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the pairs. Fill in the missing numbers.

Answer: 6 counters: 0 left over; 7 counters: 1 left over

6 counters make 3 pairs with none left over. 7 counters make 3 pairs and 1 counter is left over.

Watch for: Writing the number of pairs instead of the number left over.; Not noticing the single counter at the end.

S2. Pair up counters or draw dots in pairs. Circle Odd or Even for each number.

Answer: 4 even; 9 odd; 10 even; 13 odd

4 makes 2 pairs: even. 9 makes 4 pairs and 1 left over: odd. 10 makes 5 pairs: even. 13 makes 6 pairs and 1 left over: odd.

Watch for: Calling a big number even because it is big.; Looking at the first digit instead of the last digit.

S3. Count in 2s. Colour each number you say. Then fill in the missing number.

Answer: The last digits are 2, 4, 6, 8 and 0

The numbers are 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. The last digits are 2, 4, 6, 8, 0, 2, 4, 6, 8, 0. The missing digit is 0.

Watch for: Colouring every number.; Skipping 10 or 20.

S4. Which number is even? Write its number in the bracket.

Answer: (3) 14

7 is odd (last digit 7). 11 is odd (last digit 1). 14 is even (last digit 4). 15 is odd (last digit 5). So the even number is 14.

Watch for: Choosing the biggest number.; Choosing 11 because it has two same digits.

S5. Look at the 11 counters in pairs. Fill in the blank. Then circle Odd or Even.

Answer: 1 counter has no partner; 11 is odd

11 counters make 5 pairs. 1 counter is left without a partner, so 11 is odd.

Watch for: Writing 5 (the number of pairs).; Writing 0.

S6. Read each statement. Decide if it is True or False.

Answer: True, False, True, False

6 is even: True. 9 is odd, not even: False. 15 is odd: True. 20 is even, not odd: False.

Watch for: Marking 9 is even as true.; Thinking 20 is odd because it is big.

Think harder (page 3)

T1. Mei says: an odd number plus an even number is always odd. Test it. Fill in the blanks.

Answer: Totals 11, 13, 9, all odd; Yes, the claim works for these three

6 + 5 = 11 (odd). 10 + 3 = 13 (odd). 8 + 1 = 9 (odd). All three totals are odd, so the claim works for these three sums. Three tests are good evidence, not a proof.

Watch for: Saying Yes means the claim is true for every number.; Adding wrongly (6 + 5 = 12).

T2. Circle Odd or Even for each number.

Answer: 57 odd; 80 even; 34 even; 91 odd

57 ends in 7: odd. 80 ends in 0: even. 34 ends in 4: even. 91 ends in 1: odd.

Watch for: Looking at the first digit (the tens).; Thinking 0 is odd.

T3. Ravi says: an odd number plus an odd number is always odd. Test it with 9 + 3. Fill in the blanks.

Answer: 9 + 3 = 12; No, the claim is not true; odd + odd is Never odd (pairs picture)

9 + 3 = 12, and 12 is even. So Ravi is not right. Each odd number has one counter left over, and the two left-over counters make a pair, so odd plus odd is never odd.

Watch for: Writing 11 or 13.; Saying Yes because 9 and 3 are both odd.

T4. Can you make each number by adding two odd numbers? Circle Yes or No.

Answer: 12 yes (5 + 7); 15 no; 20 yes (9 + 11)

12 = 5 + 7 (yes). 15 is odd, and two odd numbers always make an even total, so no. 20 = 9 + 11 (yes).

Watch for: Trying only one pair for 15 and giving up too soon.; Choosing Yes for every number.

Go further (page 4)

G1. Look at 3 + 5. Draw 3 counters and 5 counters in pairs. Fill in the missing numbers.

Answer: 1 left over, 1 left over, 1 pair

3 counters make 1 pair and 1 left over. 5 counters make 2 pairs and 1 left over. The 2 left-over counters make 1 new pair, so 3 + 5 has none left over.

Watch for: Forgetting that the two left-over counters pair up.; Drawing the wrong number of counters.

G2. Add three odd numbers. Circle Odd or Even for each total.

Answer: 3 + 5 + 7 = 15 odd; 1 + 1 + 1 = 3 odd; 9 + 3 + 5 = 17 odd; odd + odd + odd is Odd

3 + 5 = 8 (even), 8 + 7 = 15 (odd). 1 + 1 = 2, 2 + 1 = 3. 9 + 3 = 12, 12 + 5 = 17. Odd + odd is even, and even + odd is odd, so the total is odd.

Watch for: Saying even because odd + odd is even.; Adding wrongly.

G3. Look at 4 + 6. Draw 4 counters and 6 counters in pairs. Fill in the blank, then finish the sentence.

Answer: 0 counters are left over; even + even is even because no counter is left over

4 makes 2 pairs and 6 makes 3 pairs. Neither has a counter left over, so together they make 5 pairs and none is left over. The total, 10, is even.

Watch for: Counting 4 + 6 = 10 without looking at the pairs.; Writing odd.

Investigation (page 5): Numbers next to each other

Always odd: 7, 11, 17, 21. One of the two numbers is odd and one is even, and odd + even is odd. Another way: a number plus the next number is a double plus 1, and a double has no counter left over, so one more is odd.

Look for: A child who tries several pairs, records odd or even, and says always. Credit any clear reason: one number is odd and one is even, or it is a double and 1 more (from last lesson, if the child was there). Do not mark a child down for stopping at a few tests.

Exit ticket (page 5)

E1. Is 17 odd or even? Circle Odd or Even.

Answer: Odd

17 makes 8 pairs and 1 counter is left over. The last digit is 7: odd.

Watch for: Choosing Even because 17 is a big number.; Looking at the tens digit.

E2. Which number is even? Write its number in the bracket.

Answer: (3) 18

9, 13 and 21 end in odd digits. 18 ends in 8: even.

Watch for: Choosing 21.; Choosing the first option.

E3. Can you make 14 by adding two odd numbers? Circle Yes or No.

Answer: Yes

7 + 7 = 14 (also 5 + 9). 14 is even, and odd plus odd is even.

Watch for: Saying No because 14 is even.; Not trying 7 + 7.

Sneak-peek puzzles (not in the workbook)

1. 1
2. No. 5 + 7 = 12 counters make 6 pairs. The one left over from the 5 and the one left over from the 7 make a pair.

Answer key: Lesson 5, Equal Groups

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Are the groups equal? Circle Yes or No for each picture.

Answer: Left: Yes. Right: No.

The left picture has 3 bags with 4 sweets in each bag: the groups are equal. The right picture has 2, 3 and 2 cookies: the groups are not equal.

Watch for: Counting only the number of groups.; Saying Yes because the pictures look alike.

S2. Look at the picture. Fill in the blanks.

Answer: 3 plates; 5 cookies on each plate; 3 groups of 5 = 15

There are 3 plates and each plate has 5 cookies. 5 + 5 + 5 = 15, so 3 groups of 5 = 15.

Watch for: Writing 5 groups of 3.; Counting the cookies on every plate and losing count.

S3. Mei has 5 jars. Each jar has 2 sweets. How many sweets does Mei have in all?

Answer: 2 + 2 + 2 + 2 + 2 = 10; 5 groups of 2 = 10

5 jars with 2 sweets each: 2 + 2 + 2 + 2 + 2 = 10. So 5 groups of 2 = 10 and Mei has 10 sweets in all.

Watch for: Adding 5 + 2 = 7.; Losing count of the jars.

S4. Draw 4 dots on each plate. Then fill in the blank.

Answer: 3 groups of 4 = 12

3 plates with 4 dots on each: 4 + 4 + 4 = 12, so 3 groups of 4 = 12.

Watch for: Drawing 3 dots on 4 plates.; Drawing different numbers of dots on the plates.

S5. Fill in the table.

Answer: 15, 8, 10

3 groups of 5 = 5 + 5 + 5 = 15. 4 groups of 2 = 2 + 2 + 2 + 2 = 8. 2 groups of 5 = 5 + 5 = 10.

Watch for: Adding the two numbers: 3 + 5 = 8.; Writing the number of groups in the Total box.

S6. Aisha has 4 boxes. Each box has 5 pencils. How many pencils does Aisha have in all?

Answer: 4 × 5 = 20 (5 × 4 = 20 is also right); 20 pencils

4 boxes with 5 pencils in each box. 5 + 5 + 5 + 5 = 20. The number sentence is 4 × 5 = 20 and Aisha has 20 pencils.

Watch for: Writing + in the circle.; Writing 4 × 5 = 9.

Think harder (page 3)

T1. Look at the dots in rows. Fill in the blanks.

Answer: 2 × 8 = 16; 8 × 2 = 16; Yes. (In (a) 8 × 2 = 16 is also right if the child counted the dots in one row first; then (b) is 2 × 8.)

The picture has 2 rows with 8 dots in each row: 2 × 8 = 16. Turned on its side it has 8 rows with 2 dots in each row: 8 × 2 = 16. The totals are the same, so 2 × 8 = 8 × 2.

Watch for: Adding 2 + 8 = 10.; Thinking the totals must be different because the pictures look different.

T2. Each plate should have 3 cookies. Look at the picture. How many more cookies are needed to make the groups equal?

Answer: 3 more cookies

The plates have 3, 1 and 2 cookies. The second plate needs 2 more and the third plate needs 1 more. 2 + 1 = 3 more cookies.

Watch for: Writing 6 (the number of cookies now).; Adding only the missing cookies of one plate.

T3. Kumar has 3 plates with 4 apples on each plate. He says he has 7 apples.

Answer: 3 groups of 4 = 12; No

3 groups of 4 apples: 4 + 4 + 4 = 12. 7 is 3 + 4, which is not the same as 3 groups of 4. So Kumar is not correct.

Watch for: Saying Yes because 3 + 4 = 7.; Writing 3 × 4 = 34.

T4. Jenny has 12 stickers. She puts them into equal groups in different ways. Fill in the table.

Answer: 3, 3

4 groups of 3 = 12, so each of the 4 groups has 3. 3 groups of 4 = 12, so the number of groups is 3.

Watch for: Writing the total in every box.; Choosing numbers that do not make 12.

Go further (page 4)

G1. A shed has 3 bikes and 4 cars. Each bike has 2 wheels. Each car has 4 wheels. How many wheels are there in all?

Answer: 6 wheels, 16 wheels, 22 wheels

3 bikes have 2 + 2 + 2 = 6 wheels. 4 cars have 4 + 4 + 4 + 4 = 16 wheels. 6 + 16 = 22 wheels in all.

Watch for: Counting a car as 2 wheels.; Forgetting to add the wheels of the bikes and the cars.

G2. Jamie has 6 jars. Each jar has 5 marbles. Jamie needs 40 marbles. How many more marbles does he need?

Answer: 6 × 5 = 30; 10 more marbles

6 jars with 5 marbles in each jar: 5 + 5 + 5 + 5 + 5 + 5 = 30 marbles. 40 - 30 = 10 more marbles.

Watch for: Writing 40 - 6.; Stopping after finding 30.

G3. Wei has 24 sweets. Show 3 different ways to put them into equal groups. Use a different number of groups each time. Many answers are right.

Answer: For example 2 groups of 12, 3 groups of 8, 4 groups of 6 (also 6 of 4, 8 of 3, 12 of 2, 1 of 24, 24 of 1)

2 × 12 = 24, 3 × 8 = 24, 4 × 6 = 24. Turning each one gives 12 × 2, 8 × 3 and 6 × 4.

Watch for: Choosing numbers that do not make 24.; Writing the same way twice.

Investigation (page 5): Rows with 12 counters

Six ways: 1 × 12, 2 × 6, 3 × 4, 4 × 3, 6 × 2 and 12 × 1. Turning a picture gives a second one (2 × 6 and 6 × 2). 5 rows does not work: 5 rows of 2 uses 10 counters and 2 are left over.

Look for: A child who builds several arrays, writes a × sentence for each and notices that turning one gives another. Credit a child who tries 5 rows and finds counters left over. Do not mark a child down for finding only three ways.

Exit ticket (page 5)

E1. Fill in the blank.

Answer: 20

4 fives: 5 + 5 + 5 + 5 = 20, so 4 groups of 5 = 20.

Watch for: Writing 9 (5 + 4).; Losing count of the fives.

E2. Which is the same as 4 groups of 3? Write its number in the bracket.

Answer: (2) 3 + 3 + 3 + 3

4 groups of 3 means 3 four times: 3 + 3 + 3 + 3.

Watch for: Choosing 4 + 3.; Choosing 3 + 3 + 3.

E3. Ali has 3 boxes. Each box has 4 pencils. How many pencils does Ali have in all?

Answer: 3 × 4 = 12 (4 × 3 = 12 is also right); 12 pencils

3 boxes with 4 pencils in each box: 4 + 4 + 4 = 12. The sign in the circle is ×.

Watch for: Writing + in the circle.; Writing 3 + 4 = 7.

Sneak-peek puzzles (not in the workbook)

1. 6
2. Same. Each of them has 12 sweets: 4 groups of 3 and 3 groups of 4 are both 12.